

FINAL PROJECT REPORT FOR MACHINE LEARNING

TOPIC:

ZERO-SHOT OBJECT COUNTING USING CLASSICAL CV, PCA/WHITENING AND FAISS

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Abstract

Zero-Shot Object Counting (ZSC) requires a system to count objects from a new category using only a few exemplar regions provided by the user. In this project, we follow a fully classical approach: each image patch is described by HOG, HSV color histograms, and LBP, then compressed with PCA/Whitening, L2-normalized, and searched with FAISS using cosine similarity. Candidate boxes are filtered by Non-Maximum Suppression (NMS), and raw counts are corrected by a small linear calibration model fitted on validation data.

The goal is not to outperform modern deep learning systems, but to show that a transparent and interpretable classical pipeline can still handle a challenging computer vision task under strict no-neural-network constraints.

1 INTRODUCTION

Object counting in images is a fundamental Computer Vision problem. Traditional counting models usually require labeled data for each target class, while Zero-Shot Counting allows the user to specify a few exemplar boxes and then searches for visually similar regions in the same image.

Most high-performing recent methods rely on CNNs, Transformers, or other pretrained deep learning models. So this approach mainly relies on the core ideas of Machine Learning and Computer Vision: hand-crafted feature design, dimensionality reduction, vector normalization, similarity search, and geometric post-processing.

The project makes three contributions:

1. It builds a classical feature representation using HOG, HSV histograms, and LBP to capture object shape, color, and texture.
2. It replaces the deep embedding model with PCA/Whitening, which reduces noise, balances feature scales, and produces embeddings suitable for cosine similarity.
3. It uses FAISS to retrieve image patches similar to the exemplar, then applies thresholding, NMS, and linear calibration to convert candidate boxes into the final count.

2 METHODOLOGY

The system contains four modules: feature extraction, PCA/Whitening, FAISS-based vector search, and count filtering/calibration. The full pipeline is shown in Figure 1.

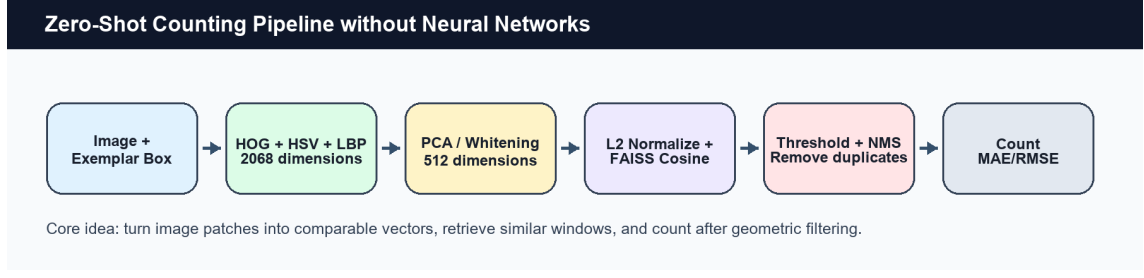


Figure 1: Overall pipeline of the classical-only Zero-Shot Counting system.

2.1 Multi-modal Feature Engineering

Because no CNN is used to learn features from pixels, the patch representation is designed manually. Each patch is resized to 64×64 and represented by:

- **HOG (Histogram of Oriented Gradients):** captures shape, edges, and gradient orientation. In the current configuration, HOG produces a 1764-dimensional vector [2].
- **HSV Color Histogram:** captures color distribution in the Hue, Saturation and Value channels. Each channel uses 16 bins, producing 48 dimensions in total.
- **LBP (Local Binary Pattern):** captures local texture by comparing each pixel with its neighbors, producing a 256-dimensional histogram [3].

After concatenation, each patch becomes a 2068-dimensional vector: HOG captures geometry, HSV preserves color, and LBP captures local texture.

Classical feature vector: 2068 dimensions

Each 64×64 patch is described by shape, color, and texture before PCA compression.



Without CNN-based feature learning, patch representation must be hand-crafted and interpretable.

Figure 2: Structure of the HOG + HSV + LBP feature vector before PCA/Whitening.

2.2 PCA/Whitening

If the original 2068-dimensional vector is used directly, many dimensions may be redundant or poorly scaled. PCA finds the main directions of variation, while Whitening balances the principal dimensions in variance [4].

For a feature vector x , the system first subtracts the feature mean:

$$x_c = x - \mu \quad (1)$$

Then it projects the centered feature onto the top k principal components:

$$z = W_k^T x_c \quad (2)$$

Whitening is applied dimension by dimension:

$$\hat{z}_i = \frac{z_i}{\sqrt{\lambda_i + \epsilon}} \quad (3)$$

Finally, the vector is L2-normalized:

$$e = \frac{\hat{z}}{\|\hat{z}\|_2 + 10^{-8}} \quad (4)$$

With this normalization, the dot product between two embeddings is equivalent to cosine similarity, so FAISS can compare patches directly in the PCA/Whitening space.

2.3 Inference: Multi-scale Window and FAISS Search

During inference, the user provides one or more exemplar boxes. Their patches are transformed through the same feature and PCA/Whitening pipeline, then averaged to form the query vector.

The image is then scanned using multi-scale sliding windows. The base window size is estimated from the exemplars and scaled by 0.8, 1.0, and 1.2. Each window becomes a database embedding through the same pipeline.

FAISS uses IndexFlatIP to retrieve the candidates with the highest inner product scores with respect to the query. Since all embeddings are L2-normalized, the FAISS score is exactly cosine similarity [5]:

$$s(q, c) = q^T c = \cos(q, c) \quad (5)$$

Candidate windows whose scores exceed the `similarity_threshold` are retained for geometric post-processing.

2.4 Thresholding, NMS, and Count Calibration

After FAISS retrieval, several windows may still cover the same object. NMS keeps high-score boxes and removes boxes with excessive IoU overlap [6].

In addition, a classical validation-time calibration step is added. For each pair of similarity and NMS thresholds, the system computes the raw count, then fits a small linear correction:

$$\text{count}_{final} = \alpha \cdot \text{count}_{raw} + \beta \quad (6)$$

The parameters α and β are fitted with light ridge regression. This step only corrects counting bias; it does not learn visual features.

3 EXPERIMENTS

3.1 Dataset setup and PCA training

The dataset used in this project is FSC-147, which contains images with diverse object categories and point annotations used to compute ground-truth counts [1]. The dataset is split into `train`, `val`, `test`, and `test_coco`.

A key point is that `train.py` loads the extracted exemplar features, fits PCA/Whitening on the training features, and evaluates several PCA dimensionalities using a validation proxy.

The main settings are:

- Feature dimension: 2068.
- PCA candidates: 64, 128, 256, 512.
- Number of training features in the checkpoint: 11079.
- Number of validation features in the checkpoint: 3946.
- Best PCA dimension: 512.

3.2 PCA dimensionality selection

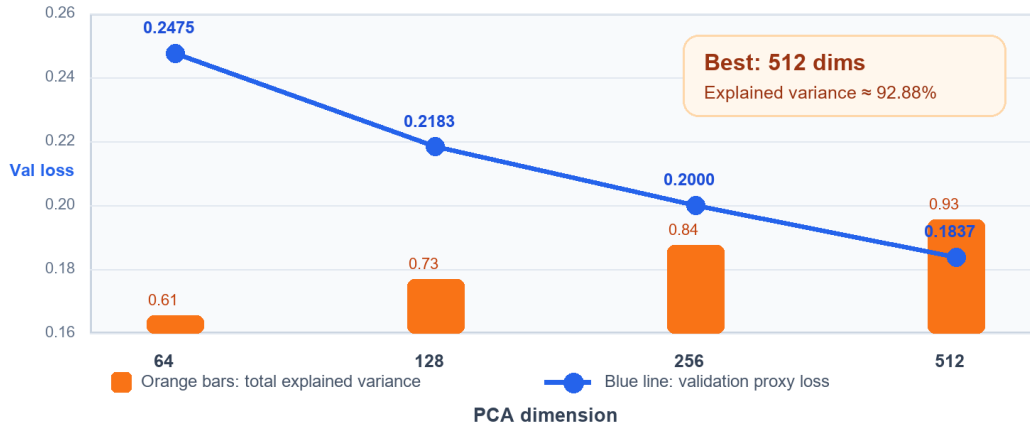
Table 1 reports the validation proxy results for different PCA dimensionalities. This loss is not the final counting MAE; it is only a proxy used to select an embedding space that better groups exemplars from the same image.

Table 1: PCA/Whitening dimensionality selection on the validation proxy.

PCA dim	Val loss	Val accuracy	Explained variance
64	0.2475	0.2446	0.6093
128	0.2183	0.2557	0.7307
256	0.2000	0.2506	0.8425
512	0.1837	0.2258	0.9288

Selecting PCA/Whitening dimensionality

Validation proxy loss decreases as dimensionality increases; the current checkpoint selects PCA dim = 512.

**Figure 3:** Validation proxy loss decreases as PCA dimensionality increases; the best checkpoint uses 512 dimensions.

3.3 Quantitative evaluation

The model is evaluated using MAE and RMSE. MAE measures average absolute counting error, while RMSE is more sensitive to large outliers. The MSE values are computed as $MSE = RMSE^2$.

Table 2: Counting evaluation results on FSC-147 after calibration.

Dataset	MAE	RMSE	Note
Test	38.38	142.79	After calibration
Test COCO	38.69	73.29	After calibration

3.4 Effect of count calibration

Calibration is added after the raw NMS count and only learns a simple linear correction from validation counts. Table 3 compares errors before and after this correction.

Table 3: Before/after comparison of count calibration. MSE is computed from RMSE.

Dataset	Setting	MAE	RMSE	MSE
Test	Before calibration	50.01	83.10	6905.61
Test	After calibration	38.38	142.79	20388.98
Test COCO	Before calibration	54.77	98.59	9720.99
Test COCO	After calibration	38.69	73.29	5371.42

Calibration improves MAE on both splits: from 50.01 to 38.38 on Test and from 54.77 to 38.69 on Test COCO. However, RMSE/MSE increase on the regular Test split, indicating that a few difficult outlier images receive larger errors after calibration. Therefore, calibration reduces average counting bias but does not fully solve extreme cases.

3.5 Qualitative evaluation

NMS is essential because sliding windows often create multiple boxes around the same object. Figure 4 illustrates the result before and after NMS.

Non-Maximum Suppression: count each object once

Many sliding windows may cover the same object; NMS keeps the high-score box and suppresses overlapping ones.

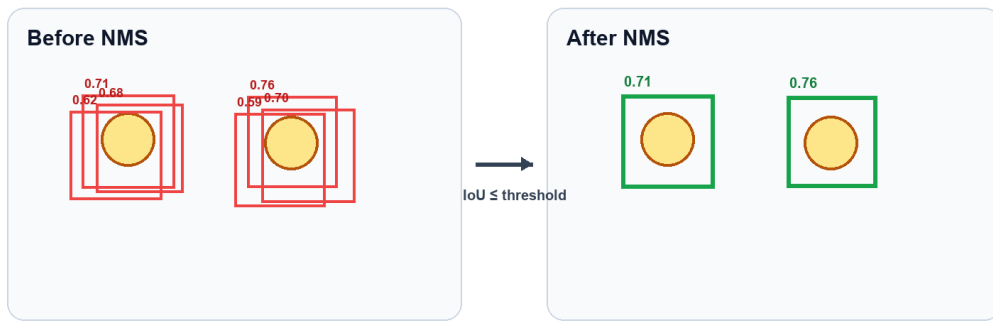


Illustration of duplicate-box suppression, not an actual FSC-147 image.

Figure 4: Illustration of how NMS removes overlapping bounding boxes.

4 DISCUSSION

Advantages. The main strength of the pipeline is transparency. Each component has a clear role: HOG captures edges, HSV preserves color, LBP captures texture, PCA reduces noise, FAISS retrieves similar patches, and NMS removes duplicate detections.

Suitability for the course constraint. Since the method does not use CNNs, Transformers, DNNs, or neural networks, it satisfies the classical-only requirement. The main learned artifact is a PCA/Whitening .npz file, not a neural network checkpoint.

Limitations. Sliding windows are the main bottleneck because many patches must be evaluated on CPU. Hand-crafted features also lack deep semantic context, so the method can fail in textured backgrounds or under occlusion.

5 CONCLUSION

This project builds a classical-only Zero-Shot Object Counting system: HOG + HSV + LBP describe image patches, PCA/Whitening compresses the embedding space, FAISS retrieves candidates by cosine similarity, NMS removes duplicate boxes, and linear calibration corrects the final count. Experiments on FSC-147 show calibrated MAE around 38 on both Test and Test COCO. The value of the project lies in interpretability, compliance with the no-neural-network constraint, and the integration of classical Machine Learning techniques into a complete working system.

References

- [1] V. Ranjan, U. Sharma, T. Nguyen, and M. Hoai, “Learning To Count Everything,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.
- [2] N. Dalal and B. Triggs, “Histograms of Oriented Gradients for Human Detection,” in *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR)*, 2005, pp. 886–893.
- [3] T. Ojala, M. Pietikainen, and T. Maenpaa, “Multiresolution Gray-Scale and Rotation Invariant Texture Classification with Local Binary Patterns,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 24, no. 7, pp. 971–987, 2002.
- [4] I. T. Jolliffe, *Principal Component Analysis*, 2nd ed. New York: Springer, 2002.
- [5] J. Johnson, M. Douze, and H. Jegou, “Billion-Scale Similarity Search with GPUs,” *IEEE Transactions on Big Data*, vol. 7, no. 3, pp. 535–547, 2021.

- [6] P. F. Felzenszwalb, R. B. Girshick, D. McAllester, and D. Ramanan, “Object Detection with Discriminatively Trained Part-Based Models,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 32, no. 9, pp. 1627–1645, 2010.